Assignment 2 The Cost of Being Wrong

Assignment	Assignment 2 : The Cost of Being Wrong
Topic	Classical ML, Careful Evaluation & Ensemble Methods
Weight	5% of your final grade
Released	Week 4 (17 Aug)
Due	Week 6 (31 Aug), 23:59 EAT
Maps to	CLO1, CLO2
Repository	a2-models

The big idea (in plain words)

Only about **4 out of every 100 borrowers actually default** that's the "4% base rate." On top of that, the two kinds of mistakes are **not equally costly**. Missing a real default (letting a bad loan through) hurts you about **12.5 times more** than wrongly rejecting a good borrower. Because of this, the model with the highest **accuracy** is usually **not** the model you should actually put into production. And the usual cut-off of **0.5** for deciding "approve or reject" is almost never the right one either.

So, this assignment is **not really about building models**. It's about **making a smart decision with them** and then writing up that decision clearly enough that a **bank regulator** would accept it.

What you'll learn to do

- Pick and defend the right way to test a model when the data is imbalanced and ordered by time.
- Tell the difference between how well a model ranks risk overall and how well it performs at one specific cut-off and choose that cut-off based on cost, not tradition.
- Build and compare two families of ensemble models (bagging and gradient boosting), and understand why they behave differently in terms of bias and variance.
- Run a fair, budget-limited hyperparameter search and report your tuning honestly.
- Check whether your model's probabilities are trustworthy (calibration), check fairness across groups, and explain predictions using SHAP.

Which data to use

- **Use the output from Assignment 1, exactly as it was.** Don't change it.
- Scored badly on A1? **You may use the instructor's reference pipeline instead no penalty.** Just say clearly in your report that you did this.
- **Do NOT fix the imbalance by deleting rows from the majority class.** That's not allowed.

What you actually have to do

1. Set up a fair testing plan

Use **blocked / purged time-series cross-validation** (this respects the time order and stops the model from "peeking" at the future). Report **PR-AUC** as your main scoring metric, and **explain why ROC-AUC is misleading** when only 4% of cases are defaults.

2. Train a ladder of models

Train at least these three, and compare them all on the exact same folds:

- Regularised logistic regression
- A random forest
- One gradient-boosting model (XGBoost or LightGBM)

3. Choose the cut-off by cost, not by habit

Instead of using 0.5, find the threshold that gives the **lowest total cost**. Use the A1 cost values: a missed default costs **KES 10,000** and a wrongly rejected good borrower costs **KES 800**. Then show how much money your cost-based cut-off saves **per 1,000 applications** compared to the naive 0.5 cut-off. (The starter code for this is provided in the original brief.)

4. Handle the imbalance the right way

Compare **at least two** approaches (for example class weighting, a SMOTE variant, or focal-style reweighting). Do any resampling **inside each fold** and **prove** that resampling outside the folds inflates your score unfairly.

5. Tune with a budget

Run an **Optuna study of at least 60 trials**. Report your **search space**, your **pruning rule**, and the **optimization history**.

6. Check calibration

Draw a **reliability diagram** and compute the **Brier score**. Apply **Platt or isotonic calibration**, then **recalculate your cost-based cut-off** on the calibrated scores. Say whether calibration moved the threshold, and why.

7. Explain and check fairness

Produce **global and local SHAP explanations**. Report the **approval rate** and **false-negative rate** broken down across at least **two sensible subgroups** (e.g. tenure band or region). Discuss what the **CBK Digital Credit Providers Regulations (2022)** expect regarding explaining rejections to customers.

What to hand in

A repository called **a2-models** containing:

- A model comparison table
- Calibration curves and PR curves
- Your Optuna study database

- **A 4-page report** that ends with one short paragraph recommending **one single model and one single threshold** for deployment.

How it's graded

What we grade	Marks	What we're looking for
Evaluation plan	20	You set up purged/blocked time-series cross-validation correctly and explain why your chosen metric fits a 4% default rate.
Ensemble models	20	All three model types trained and compared on the same folds, with a real discussion of bias vs. variance.
Cost-based decisions	20	You find the best threshold from the cost formula and show the money saved per 1,000 applications.
Imbalance & tuning	15	Resampling done inside each fold, leakage shown, and your tuning budget and search space reported.
Calibration	10	Reliability diagram and Brier score before and after, with the threshold recalculated.
Explainability & fairness	10	SHAP at both global and local levels, plus subgroup metrics discussed against the regulation.
Reproducibility & report	5	A fresh clone reproduces your comparison table, and the report stays within four pages.
Total	100	